Student Name: Johan Fuhri
Date Exam written: 14 November 2025
Instructor name: Riaan Struwig
Instructor Signature: _____
Date Marked: 19 June 2026

MARKS

A		B		C		D		E		F				
17	+	42	+	27	+	60	+	9	+	14	=	169	=	86.7 %
20		**50**		**34**		**67**		**9**		**15**		**195**		Pass Mark = 85%

PASS!

BASIC LICENCE THEORY TEST

Paragliding Training Manual (Version February 2006)

You are encouraged to complete this paper as soon as convenient, during or after your course, whilst all information is still fresh in your mind.
All 'airlaw' questions to be passed.

NOTE: This is an open book research examination. Please answer concisely. The test answers must be your own effort and should not be copied from others!
Draw sketches where required (distinguish between describe and/or illustrate).
Where sketches are provided, write/draw answers on them.
The test can be completed online and then printed to finish the drawings/illustrations and hard copy to be submitted to your instructor.

SECTION A: GENERAL

1. Why use a logbook? (3)

Regulatory Compliance
Training and Experience Tracking
Safety and Accountability
Instructor and Category Ratings
Standardization

2. Accidents and Incidents:

a) What is a reportable accident/incident? (1)

Reportable Accident is any accident that results in serious injury, fatality, significant damage to the aircraft, or loss of control/emergency landing due to equipment failure or environmental conditions. A Reportable Incident is an incident that may not result in injury or damage, but still pose a significant risk, such as near misses, equipment malfunction that could have led to an accident, breach of airspace regulations, dangerous weather encounters, or any situation that could have led into an accident.

b) Why should accidents and incidents be reported? (1)

To ensure safety oversight, trend analysis and compliance with aviation law.

3. Which paragliding official(s) should be notified and the report be given to after an accident or incident? (1)

National Safety Officer of SAHPA and the Club Safety Officer.

4. State the procedures for:

a) Flying at an established Club or other site, for the first time in your flying career, and at any other time. (3)

New sites:

- Pilot must receive a comprehensive site briefing from a local instructor or experienced pilot familiar with the site.
- Pilot must ensure that they have permission to fly from the land owner or site manager, unless site is listed in the SAHPA AIP ENR 5.5 register.
- Pilot must comply with site rules
- Pilots must use the appropriate site radio frequencies and proper radio etiquette.
- Pilots must be current SAHPA members and have third-party liability insurance.

Also at any other time:

- Pilots must continue with site rules, SAHPA regulations and CAA regulations.
- Must have a valid license, rating and medical certificates.
- Comply with safety regulations and conduct

- Be aware of the weather and flying conditions.

b) Attempting to establish a new site, which has never been flown before by anybody. (4)

- Pilot must conduct a thorough safety assessment of the site.
- Get written permission from the landowner or controlling authority of the site.
- Pilot must notify SAHPA of the intent to fly from the new site.
- Should ideally be conducted under the supervision of a senior pilot or instructor.
- Pilot should submit a site report to SAHPA after flying.
- Pilot may propose the site to be registered and included in the SAHPA AIP ENR 5.5 register.

5. How does the sign-off system operate? (2)

During the training phase, all flights and exercises completed by a student pilot must be logged and signed off by a SAHPA-approved instructor. The logbook serves as an official record of training progress and is required when applying for a Basic Paragliding License or any higher rating. The instructor's signature confirms that the student has met the required standards for each task and flight, ensuring safety and regulatory compliance.

6. Why does the South African Hang Gliding & Paragliding Association exist, and what are the advantages of being a member of (SAHPA) (5)

The South African Hang Gliding and Paragliding Association (SAHPA) exists as an Approved Recreational Organisation (ARO) under the South African Civil Aviation Regulations (SA-CARs), specifically Part 149.

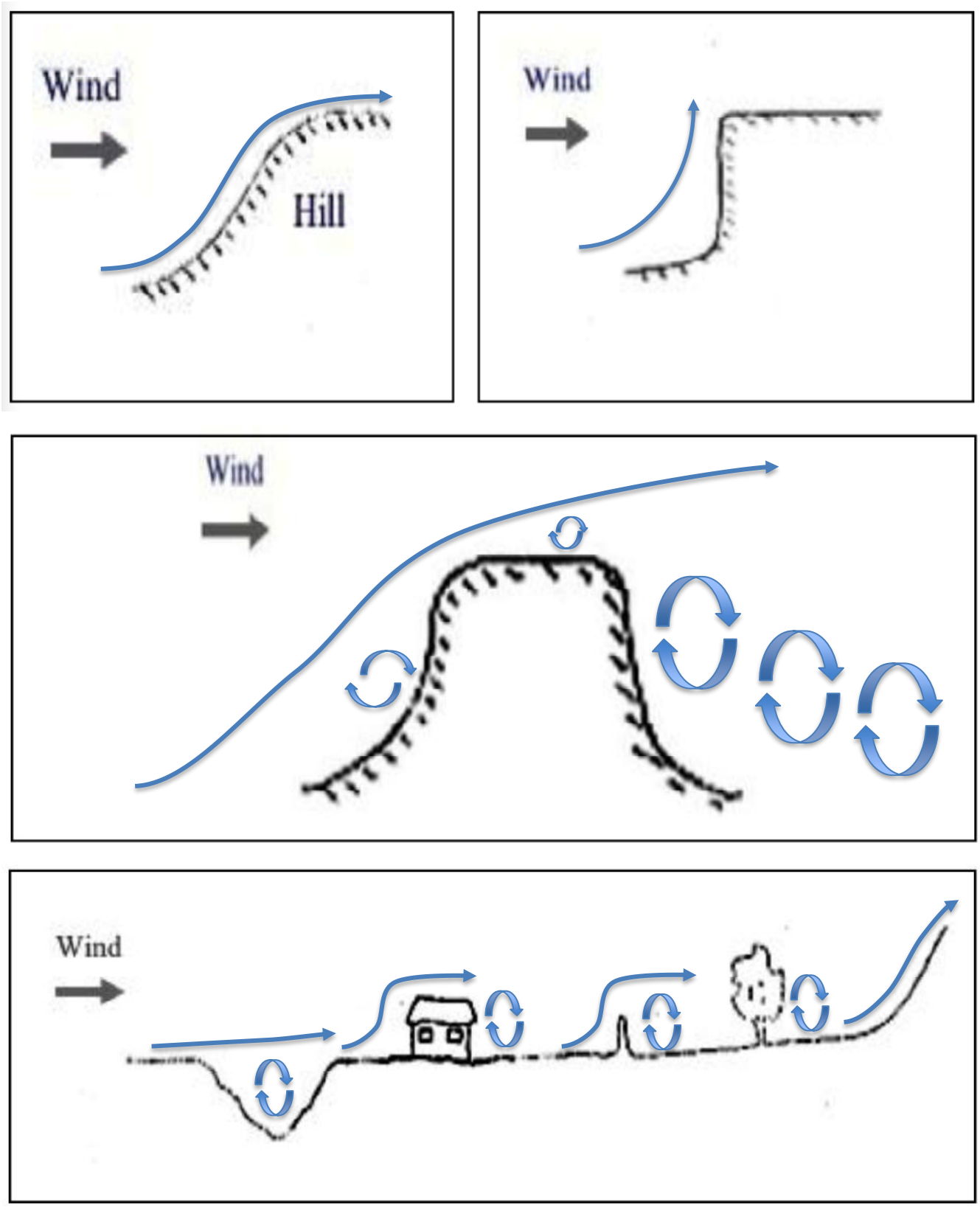
Advantages include:

- Legal and Regulatory Compliance
- Third Party Insurance
- Access to training and licensing
- Safety and oversight
- Presentation and advocacy
- Community and resources
- Disciplinary and support structures

Section Total = [20]

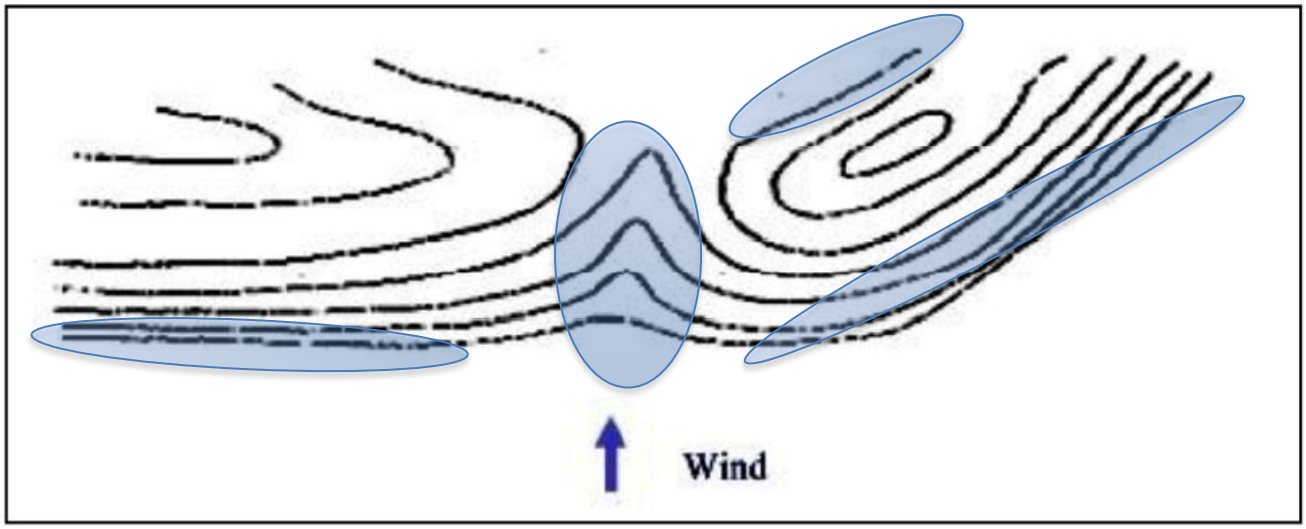
SECTION B: AIRFLOW

7. Indicate a 30 Kph wind blowing over the profiles in the illustrations below (5)

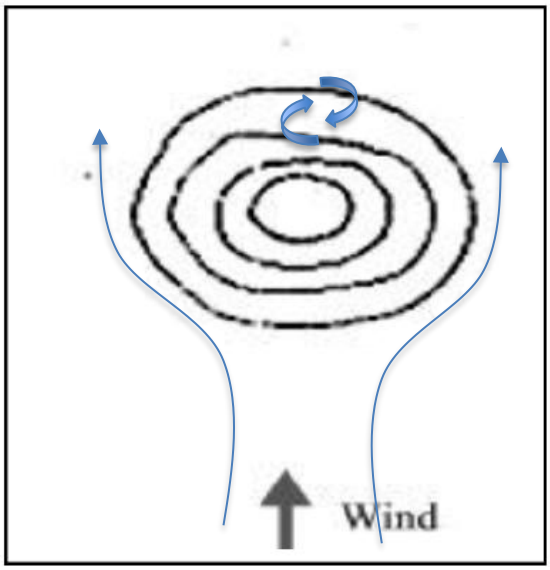


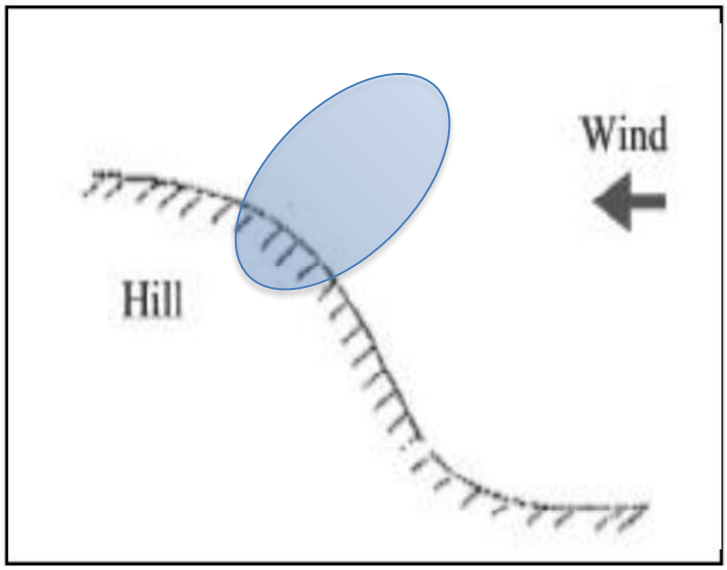
8. Contour Map

a) Indicate a 30 kph wind blowing over the contour map and shade the areas of possible penetration difficulties (3)



b) Indicate a 30 kph wind blowing over the contour map (1)





9. Shade in the lift area on the hill in the above diagram (bottom right) (1)

10. Standing at take off, how can you judge if the wind is turbulent BEFORE flying? (3)

- Time of day, turbulence is usually more prevalent during the middle of the day.
- Obstacles will cause rotor and other types of turbulence.
- Gusty conditions indicate the presence of turbulence, i.e. look at the windsock.
- Presence of cumulus clouds indicate thermals, which is associated with turbulent conditions.

11. State five visual references which could be used to assess wind SPEED

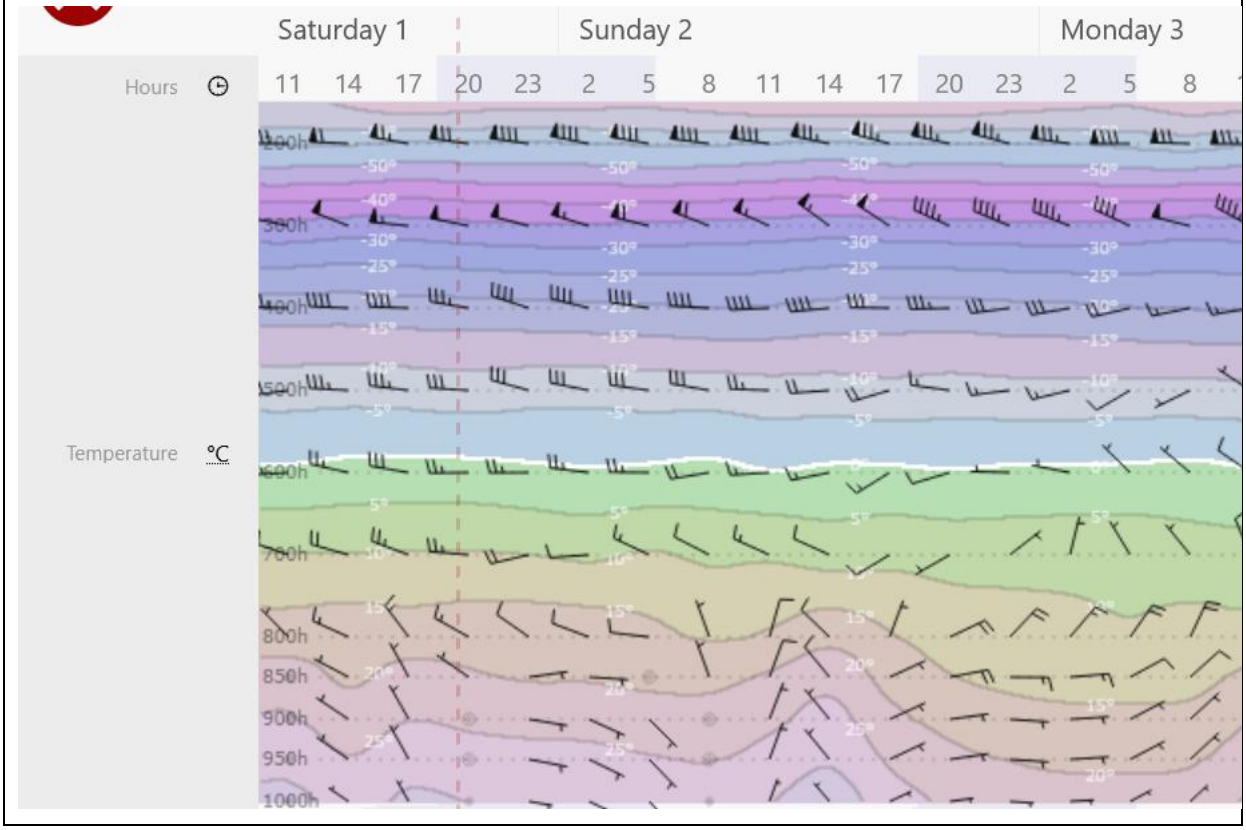
and DIRECTION. (5)

- Wind sock
- Smoke, dust or mist
- Vegetation (trees, plants)
- Cloud movement
- Wing behavior during ground handling

12. Describe (define) and illustrate wind gradient. (3)

Wind gradient refers to the change in wind speed and/or direction with altitude. Typically the wind is faster at higher altitudes.

The screenshot below, taken from Windy on 1 November 2025, shows the vertical wind gradient at Grasslands Airfield.



13. Describe and illustrate the effects of wind gradient on a paraglider while soaring above (higher than) the ridge. (Not to confused with venturi effect) (3)

When ridge soaring, above the ridge, the glider may be high enough above the ridge to a point where the into-wind ground speed is less than the overall wind speed, blowing the glider behind the ridge into an area of rotor.



14. In the above case in Q13, what safety precautions can one take? (4)

The pilot should give himself enough leeway to exercise various options like big-ears or speed bar to escape, should the wind unexpectedly increase.

15. What is a wind shadow, and where will you expect to find it? (2)

A wind shadow is a sheltered area on the downwind (leeward) side of an obstacle where the wind is significantly reduced in speed or blocked entirely. The air in this region could also become turbulent, less organized and slower.

16. Wind Shadow

a) What would the effect of wind shadow be on a paraglider, (what would it feel like and why) (2)

Flying into a wind shadow, the pilot might experience a sudden loss or airspeed, and may run the risk of a wing collapse. This is especially true if the wind shadow area is turbulent. The pilot may experience a sudden forward pitch in the wing, sudden drop (loss of lift), or shaking/jolting due to turbulence.

b) What should the pilot do when flying into wind shadow? (2)

- Anticipate and avoid: Try and prevent such situations by anticipating regions of wind shadows and staying away.
- Active flying: Keep hands on the brake toggles, and use them to feel the pressure in the wing. This assists in quick reactions to surges and collapses. Pull the brakes as necessary to keep the wing inflated in cases of wing unloading leading to leading edge collapse.
- For PPG pilots, apply throttle to increase air speed and climb away to safe altitude.

c) What should the pilot not do, and why not? (2)

Do not pull the brake too hard when suddenly surging forwards into a wind shadow, as this could

cause a stall. Leave the wing to pick up speed and settle into the appropriate speed.

17. What dangers are associated with flying in severe turbulence? (2)

- Wing collapse (deflation)
- Cascading malfunctions, like spiral dives and line twists
- Loss of altitude
- Pilot over-control leading to stalls

18. What must the pilot do when flying in turbulence? (2)

Active flying, including

- Maintain constant pressure and line tension, for best response time to changes
- Counter pitching movements (surges and lags)
- Manage asymmetric pressure and rolls (side to side movement)
- Maintain airspeed
- Prioritize altitude and landing options

19. What are the correct actions the pilot must do when

a) Turbulence causes an A-symmetric tuck, and what are the associated dangers?(4)

- Stop the rotation by using weight shift and “flying the open side”.
- Clear the collapse by pumping the collapsed side brake.
- Manage the surge by catching the forward surge once the wing reinflates.

Associated dangers:

- High speed spiral dive
- Stall/spin on the flying side
- Cravat or wingtip snag
- Low-altitude impact

b) A front tuck occurs while the speed bar is used? (2)

- Immediately release the speed bar
- Reinflate the wing, but be careful to use the brakes, as this could lead to a stall.
- Catch the surge when the wing reinflates.
- Release the brakes and fly out of the dive.

c) A full stall occurs due to deep brake application and/or turbulence.

Explain how the recovery should be made, and what should be avoided? (4)

- Full and smooth release of the brakes
- Check the wing’s status. The wing may pitch forward aggressively leading to a frontal collapse.
- Catch the surge when the wing reinflates.
- Fly out and stabilize.

Avoid:

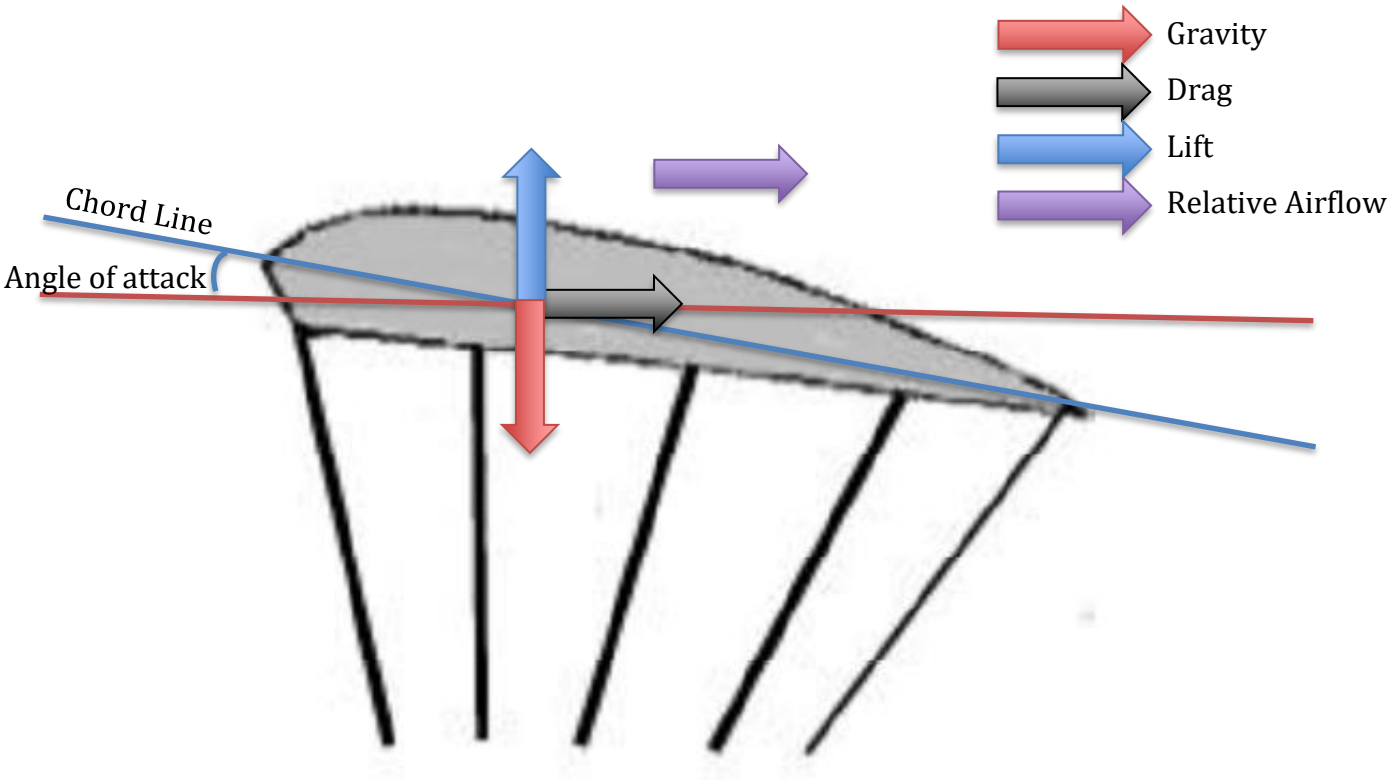
- Pumping the brakes
- Assymetric release of the brakes
- Holding the brakes too deeply
- Avoid “freezing”, and not doing anything.

Section Total = [50]

SECTION C: AERODYNAMICS

20. On the cross section sketch of the paraglider indicate the following: (6)

a) Gravity b) Drag c) Lift d) Relative Airflow e) Chord Line f) Angle of attack



21. What causes a paraglider to move forward in flight? (2)

The forward motion of a paraglider in flight is primarily caused by gravity acting on the pilot and the wing, which is constantly converted into horizontal movement via the wing's aerodynamic design.

22. Aerofoil:

a) the design of a paraglider is an aerofoil, but it is soft and formless on the ground. How is the shape formed, and maintained during flight? (2)

The paraglider wing's aerofoil shape is formed by the principle of ram-air inflation. Internal pressure, in the pockets of the wing, is created and maintained by the forward motion of the wing via the openings at the leading edge of the wing.

b) Under what circumstances can the soft wing aerofoil collapse? (2)

The wing can collapse when the wing loses its internal pressure, which can be caused by:

- External forces, like turbulence or wind shear
- Negative angle of attack (frontal collapse)
- Asymmetric disruption (asymmetric collapse)
- Too high angle of attack (stall)
- Wing malfunction

23. What happens when the brake (toggle/steering) lines of the paraglider are progressively pulled down during a flight? (3)

Pulling the brake increases the angle of attack and drag on the wing, leading to:

- Reduction in airspeed
- Temporary increase in lift
- Turning (asymmetric application of brakes)
- Stalling

24. Front Risers:

a) What is the result when the front risers are pulled down or the speed bar used, on take-off or during flight? (2)

Pulling on the A-risers decreases the angle of attack, leading to increased airspeed. It can be used during take-off to control the ascent rate and prevent wing overshoot, and can be used during flight to increase airspeed for better penetration through turbulence.
(Above according to my research on the web, although both of these seem very dubious and dangerous, in my personal opinion…)

b) What is a possible danger when doing this? (1)

Pulling on the A-risers during flight can lead to frontal collapses.

25. Define

a) Glide ratio (1)

Glide ratio is the ratio of the horizontal distance travelled to the loss of altitude while gliding unpowered.

b) Sink rate (1)

Sink rate is the amount of height lost over time, measured in m/s.

c) Wing loading (1)

The total weight of the pilot and equipment divided by the projected wing area.

26. What are the effects of

a) A light wing loading (below manufacturers specification)? (2)

Improved sink rate, but poorer forward speed and penetration. Wing becomes sluggish.

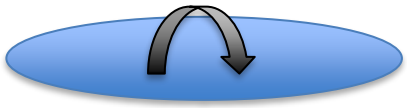
b) A heavy wing loading (above manufacturers specification)? (2)

Flying with a high wind loading will give better penetration and more resistance to collapses, but run the risk of stalls and highly aggressive recovery forces after a collapse.

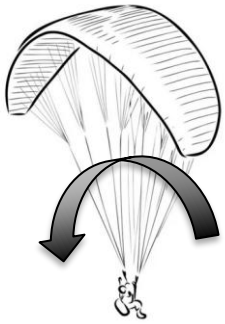
27. Illustrate:

a) Yaw (1)

Viewed from above:

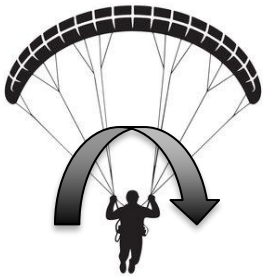


b) Pitch (1)



(Pitch forward and backward)

c) Roll (1)



(Roll left and right)

28. What would ground speed be when flying at 20 Kph airspeed in:

a) A 10 Kph headwind (1)

10 kph forward

b) A 20 Kph tailwind (1)

30 kph forward

c) A 30 Kph headwind (1)

10 kph backwards

29. When is a stall utilized to its maximum? (1)

During the flare at the end of a landing run

30. Which two features contribute to stability in paragliders**?** (2)

Aspect ratio

Wing/aerofoil profile

Section Total = [34]

SECTION D: FLIGHT & CONTROL

31. Safety:

a) Which four PRINCIPLES to check airworthiness must be followed during the daily equipment inspection? (In each of the four cases, summarise what you would check for, OR why you would do it.) (8)

- Visual: Look at material, attachment points, lines, stitching, webbing, links, etc for for signs of wear, fraying or tears.
- Physical: Pull on stitching, lines, harness attachments, straps, especially if you think they are looking unsafe.
- Systematic: Use a specific sequence of pre-day checks by starting at one point and systematically working around all equipment.
- Consistent: Always follow the same sequence and start over if interrupted.

b) When should the daily inspection be done? (2)

Before the first flight of the day and during the day if any possible damage may have occurred due to some event.

c) When should a paraglider be inspected by the manufacturer or distributor? (2)

Annually, or if there is reason to believe that the equipment is faulty or may have been compromised during an event.

32. PreFlight:

a) What are the pre-flight or pre-launch checks (before pulling up for takeoff), and what are you checking for on each one (also known as 5 point check)? (16)

- Layout: The glider must be undisturbed in the desired mushroom shape.
- Lines: Lines should be untangled, with the A-lines on top. Brake lines should not be tangled in the lines and risers.
- Connects:
    - Shoes are fastended
    - Harness connection points, including leg, waist and chest strap
    - Speedbar is connected
    - Reserve pins are fully inserted
    - Helmet is on and connected
- Wind: Speed and direction with limited gust factors
- Airspace: Clear, not pilots flying into your intended flight path.

b) What are the take-off / launch checks DURING a forward pull up? (3)

- Is the glider up and inflated properly?
- Are the lines OK?
- Is the airspace clear?
- Is your airspeed good?

33. Why is take-off and landing executed into wind? (1)

To increase the airspeed over the wing relative to the ground, less need to create speed by running.

34. What should a pilot check for when visiting a new landing area before flying? (5)

- Check that the area is obstacle free
- Not in the lee of any significant obstacles.
- Select a landing time and spot where you don't need to focus on collision avoidance with other gliders.
- Land owner permission
- Altitude, temperature and humidity

35. Arriving on site in strong wind conditions, what factors would determine whether you would be able to have a safe flight? (4)

Pilot experience
Launching conditions, wind, gusts
Is the wing fast enough
Wing loading

36. Why is it more dangerous to fly a high performance paraglider than an intermediate paraglider? (3)

- High performance wings, having higher aspect ratio, requires more active piloting to avoid asymmetric collapses and cravats.
- Intermediate wings have more passive safety, reacting less aggressively to turbulence and recover "automatically" in many cases.
- High performance wings have smaller safety margins, and may enter into deep stalls and spins more easily, and requires correct and timely inputs to recover from these.

37. 80% Brakes:

a) What is the sequence of events when 80% brakes are <u>suddenly</u> applied and <u>immediately</u> released again to zero position? (4)

- Upon pulling the brake, the wing will pull back behind the pilot, and create a temporary lift situation.
- When released, the wing will quickly surge forward and significantly in front of the pilot.
- Because of the sudden movements, the pilot speed has not significantly changed, and can cause a collapse.
- This causes a sudden dive.

b) Why can this be dangerous? (2)

The sudden surge forward can cause a sudden dive, causing the pilot to fly into the ground. In addition, if the wing surges too far forward, the wing could also get into a negative angle of attack, causing a front tuck collapse.

38. In helping another pilot during the actual launch period, what are the duties of a helper? (3)

- A helper should assist the pilot if he gets into difficulties, like getting dragged away by a gust, etc.
- Observe the pilot during the launch and inform the pilot of any issues that the pilot might not notice, like tangled lines, inflation issues, other traffic in the area, etc.
- Prevent the pilot from going over the edge, as necessary.

39. PLF:

a) What PLF stand for? (1)

Parachute landing fall

b) When is a PLF necessary? (2)

- When using a reserve parachute that is not steerable
- Stalled wing at low altitudes, where using the reserve may not be viable.

40. Reserve Parachute:

a) When should one deploy a reserve parachute? (3)

- Mid air collision at congested flying sites
- Structural failure
- Glider out of control
- Medical emergency

b) What are the steps you should follow when deploying a reserve (how)? (5)

Look, grab, look, throw, deflate.

41. What are the correct procedures to follow when one sees a paragliding accident occurring? (3)

- Immediately go to the scene while remaining aware of your own safety
- Call for help
- Provide directions to emergency staff
- Provide basic first aid
- Record details

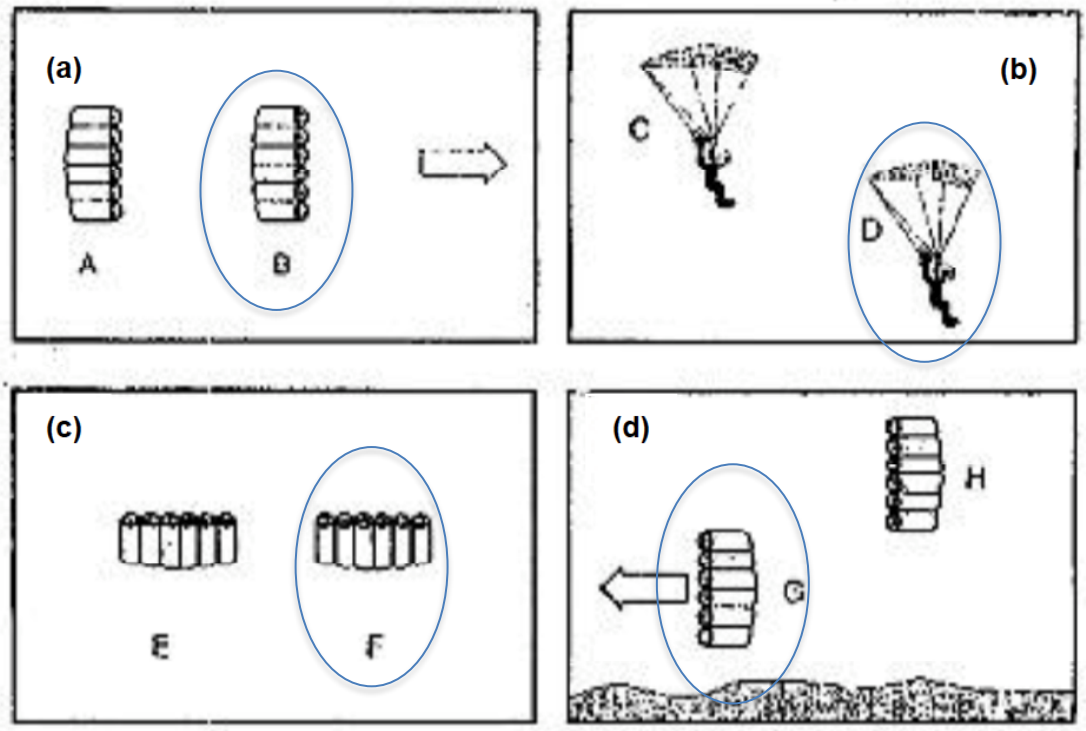
Section Total = [67]

SECTION E: AIRLAW

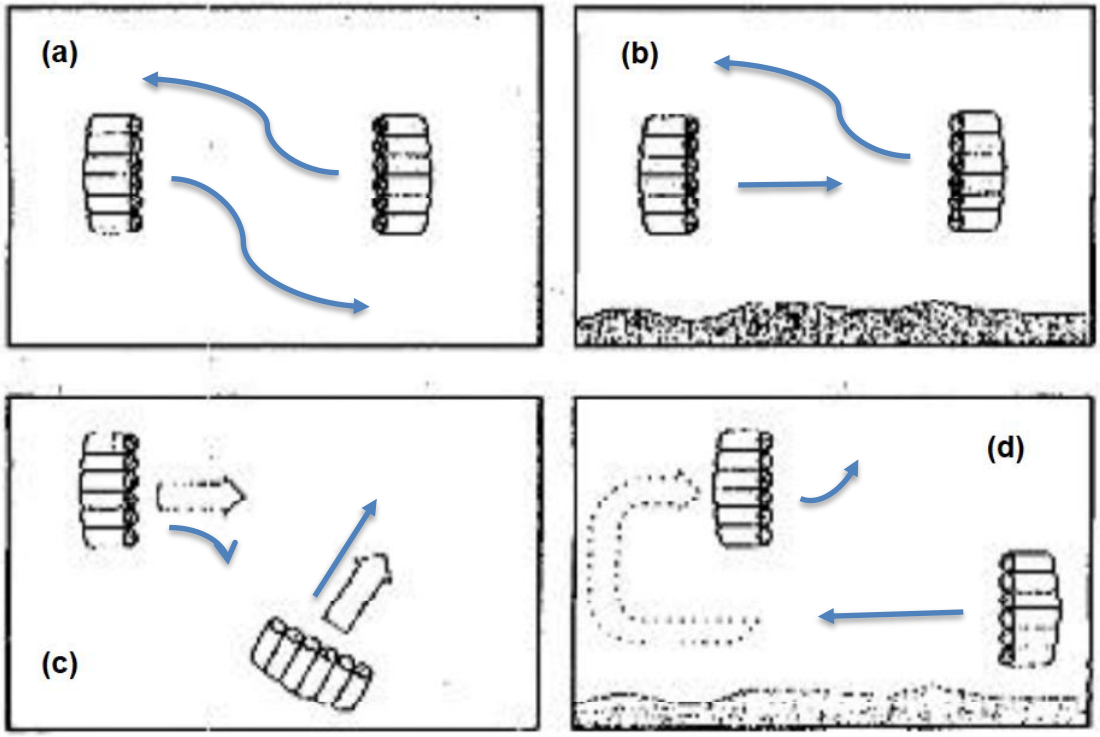
42. What is the basic rule to determine who has right of way in the air? (1)

Right is right

43. In each of the sketches on page 42, circle the glider that has right of way. (4)



44. In each in-flight situation in the sketches below use directional arrows to indicate actions to be taken by each of the paragliders in each situation. (4)



Section Total = [9]

SECTION F: COMMON SENSE

45. Irrespective of experience, why is it important to have an observer during take off? (1)

So someone can call for assistance if something goes wrong during take-off.

46. What is the danger of attempting a cliff take off? (1)

Turbulent or updraft winds near the cliff could pull the pilot off the cliff during take off, and there is not space for emergency landing in the case of aborted landing.

47. Of the two important factors of micrometeorology and glider handling ability, which is the more crucial? (1)

Micrometeorology

48. What are the dangers associated with flying in the vicinity of the following cloud types: (4)

a) Cumulonimbus

Apart from the very high winds, turbulence in a very large radius around the storm, there is execessive amounts of lift that are capable of sucking the pilot up to dangerous heigths.

b) Orographic

Since these clouds can develop very quickly, the pilot can unexpectedly be enveloped in a cloud, reducing visibility and causing disorientation.

49. When flying away on a cross country flight, what are the two most important observations that you will be making during the flight? (2)

Changing weather conditions and potential emergency landing sites

50. Why is it a rule that you attain an upright or standing posture in the harness on landing approach? (3)

To allow landing on your feet, in a running position. This is necessary to prevent injuries and damage to equipment.

51. Why would one report to the most senior pilot or site controller on a flying site prior to commencing any flying activity? (2)

To make sure your understanding of the rules is correct, and to ask for guidelines on dangerous weather conditions, landing ares issues, no-go areas , special rules, etc.

52. What are the correct actions to take in the event of an unavoidable landing on an obstacle? (1)

Don’t panic, flare as usual and simply stand into it, close your legs.

Section Total = [15]

Grand Total = [195]